

Thura Win Kyaw

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Profile

I'm an undergraduate researcher focused on post-training small Vision-Language Models (VLMs) to improve reasoning faithfulness and reduce hallucination. My current research explores reinforcement learning and preference-based post-training, including GRPO and DPO, with an emphasis on data-, label-, and memory-efficient methods. Alongside my research, I'm completing a B.E. in Electrical Engineering at Chungnam National University and work as an AI/ML Engineer Intern at AidanBio, where I build LLM/RAG systems, AI backends, and biomedical AI applications.

Research Interests

Vision-Language Models; Reinforcement Learning; Preference-Based Post-Training (GRPO, DPO); Hallucination Mitigation; Faithful and Grounded Multimodal Reasoning; Efficient Small VLMs.

Research Projects

MoonDPO – Memory-Efficient DPO for Hallucination Mitigation in Small VLMs 🌙

- Applies LoRA-based DPO to Moondream2 (1.9B) using preference pairs automatically generated from COCO annotations, without human or API labels.
- Derived and implemented the DPO objective from scratch; the full pipeline is reproducible on a 16 GB laptop.
- Preliminary (proof-of-concept run): CHAIR_i 0.135 → 0.124, CHAIR_s 0.215 → 0.200; POPE F1 essentially unchanged across splits. Caption length and objects-per-caption stay flat—the improvement is not from “saying less.” Single seed, small training budget; scaling data and running multiple seeds to establish the effect.

AidanMed – Label-Efficient VLM Post-Training for Faithful, Grounded Medical Reasoning

- Undergraduate research at AidanBio training a VLM with GRPO to produce reasoning together with a bounding box that must remain faithful to the model's own reasoning.
- Status: In progress. SFT cold-start complete; GRPO post-training is running on 8×A100 GPUs. No evaluation results reported yet.

ChatPDB – Protein Structure Visualization Assistant (LLM + RAG + PyMOL)

- Built an assistant that interprets natural language queries to visualize protein structures with LLM + RAG and PyMOL rendering.
- Implemented a retriever pipeline (BM25 + FAISS + Cross-Encoder) using UniProt annotations and enabled session-based workflows.
- Integrated FastAPI services and a CLI using local LLMs (Ollama) for interactive usage.

Experiences

AI, Software, Prompt Engineer, [AidanBio](#) – Daejeon, KR

Dec 2024 – Present

- Developed AI backend systems, APIs, model-serving infrastructure, and LLM-powered applications.
- Built RAG pipelines using vector databases, semantic search, prompt engineering, and domain-specific knowledge bases.
- Fine-tuned Vision-Language Models (VLMs) for medical AI using multimodal SFT, grounding, GRPO, and evaluation pipelines.
- Worked on knowledge distillation and teacher–student training to transfer reasoning from large multimodal models to smaller models.

AI Software Engineer Intern, [GRINDA AI](#) – Daejeon, KR

Oct 2025 – Nov 2025

- Built an AI-powered Slack bot that converts issue reports into structured GitHub issues using Claude AI and FastAPI, with auto-labeling, translation, and monitoring

Education

Chungnam National University, B.E. in Electrical Engineering

Sept 2022 – Present

- Coursework: Computer Programming (C), Computer Scientific Thinking, Linear Algebra, AI and Future Society, Modern Control Theory and Design, Sensor and Measurement Engineering, Artificial Intelligence and Data Utility

Skills

Programming Languages: Python, C, MATLAB.

VLM Research & Post-Training:

- Vision-Language Models; Hallucination Mitigation; Faithful and Grounded Multimodal Reasoning; Small VLMs
- Reinforcement Learning; SFT; GRPO; DPO; LoRA; Preference Data Generation
- POPE; CHAIR; LLM/VLM Evaluation; Hallucination and Faithfulness Analysis

Applied LLM Systems: RAG Pipelines, LangChain Orchestration, Vector Databases, Prompt Engineering, Local LLM Deployment (Ollama), FastAPI, API Development, AI Model Integration.

Tools & Platforms:

- Development: VS Code, Cursor AI, Jupyter Notebook, PyMOL
- Version Control: Git, GitHub
- Productivity: Notion, Google Workspace, Microsoft Office
- Model Hosting & Inference: Ollama, Hugging Face, Transformers Inference API

Operating Systems: Windows, Linux, Mac.

Languages: Burmese (Native), English (Fluent), Korean (TOPIK 5).

Foundational ML

- Iris Flower Classification (SVM) 🔄
- Customer Segmentation (K-means) 🔄
- Text Classification 🔄
- Quote Sentiment Analysis 🔄

References

Available upon request.